SyncVDI User Guide

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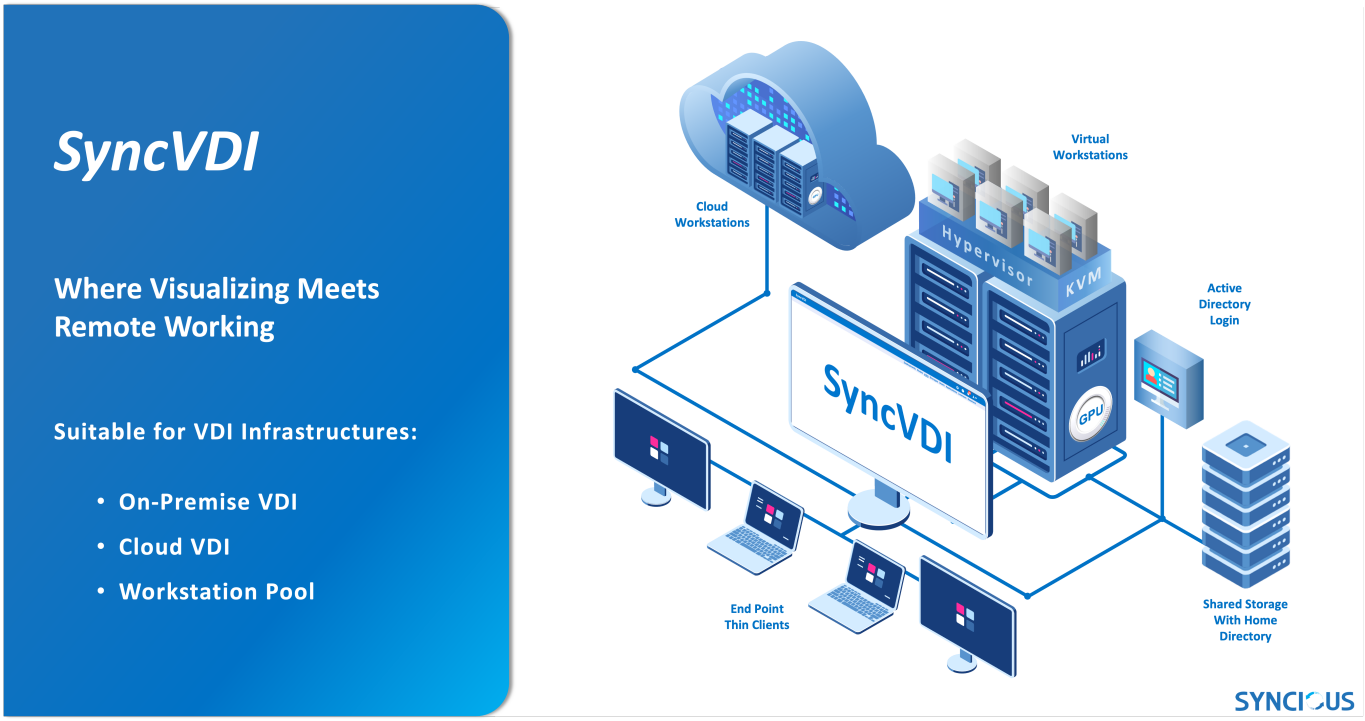
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1. Introduction of SyncVDI

SyncVDI is a management platform designed for users to access and interact with a desktop environment hosted on a remote server or data center. Instead of having their desktop environment stored locally on their physical device, users connect to a virtualized desktop instance over a network, typically using a remote display protocol. It provides a unified and centralized solution for deploying, managing, accessing, and requesting sessions for workstations, both on-premises and cloud-based resources.

In the SyncVDI environment, all user data, including documents, settings, and application data, is stored centrally on the servers or storage infrastructure in the data center or cloud. This centralized storage approach allows for better data management, backup, and recovery processes. It provides enhanced security and compliance capabilities, reducing the risk of data breaches.

SyncVDI is pluggable with existing on-premises infrastructure and cloud-based services to provide a centralized platform for VDI admins to manage, access, and monitor a pool of workstations. Admins can easily have a detailed overview of users' activity with the help of the SyncVDI dashboard, improving the overall efficiency of the workflow.

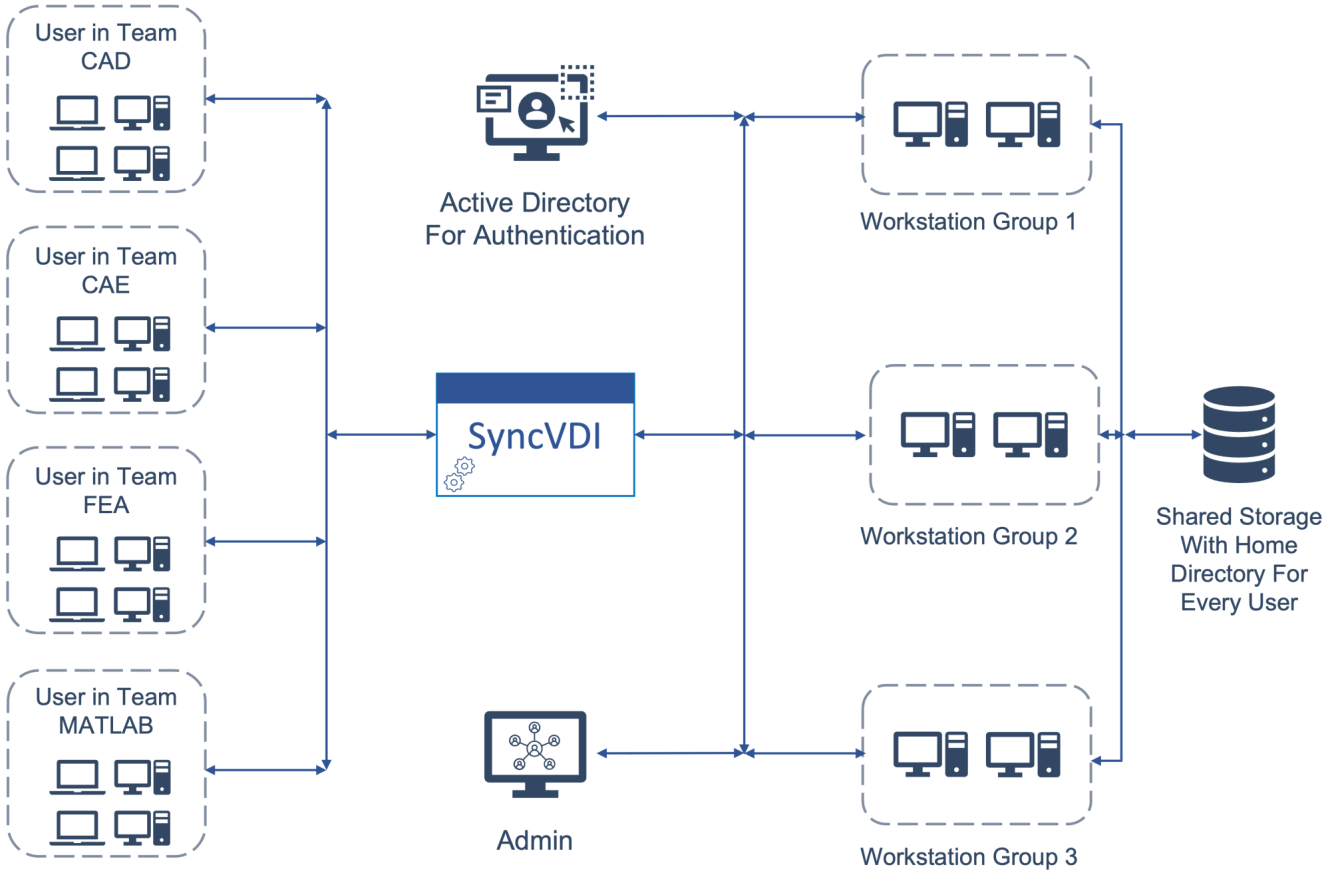


2. SyncVDI Architecture

The architecture of SyncVDI is designed to provide a centralized infrastructure for accessing virtual workstations remotely, ensuring the highest level of security, saving on CAPEX expenses, improving resource utilization, and facilitating scalability. The architecture of SyncVDI comprises several components, including:

- **Front-end:** The front-end component provides the user interface for the SyncVDI platform. This component allows administrators to manage and monitor resources, allocate resources to user groups, and manage user accounts and permissions.
- **Back-end:** The back-end component is responsible for managing and maintaining the underlying VDI infrastructure. This component includes a database for storing information about resources, user accounts, and resource allocation, as well as a set of tools for monitoring and optimizing resource utilization.
- **Resource manager:** The resource manager component is responsible for allocating and managing VDI resources, such as workstations, storage systems, user management, and network infrastructure. This component allows administrators to specify resource allocation policies and allocate resources to user groups in a controlled and optimized manner.
- **Monitoring and reporting:** The monitoring and reporting component provides real-time visibility into the performance and utilization of VDI resources. This component includes a set of tools for monitoring resource usage, performance, and availability, as well as generating reports on resource utilization and performance.
- **Security:** SyncVDI includes a comprehensive security framework that ensures the privacy and security of sensitive data and resources. This framework includes features such as authentication and authorization, encryption, and data backup and recovery.

The architecture of SyncVDI is designed to be pluggable, flexible, scalable, and easy to use, allowing administrators to effectively manage and optimize their VDI resources.



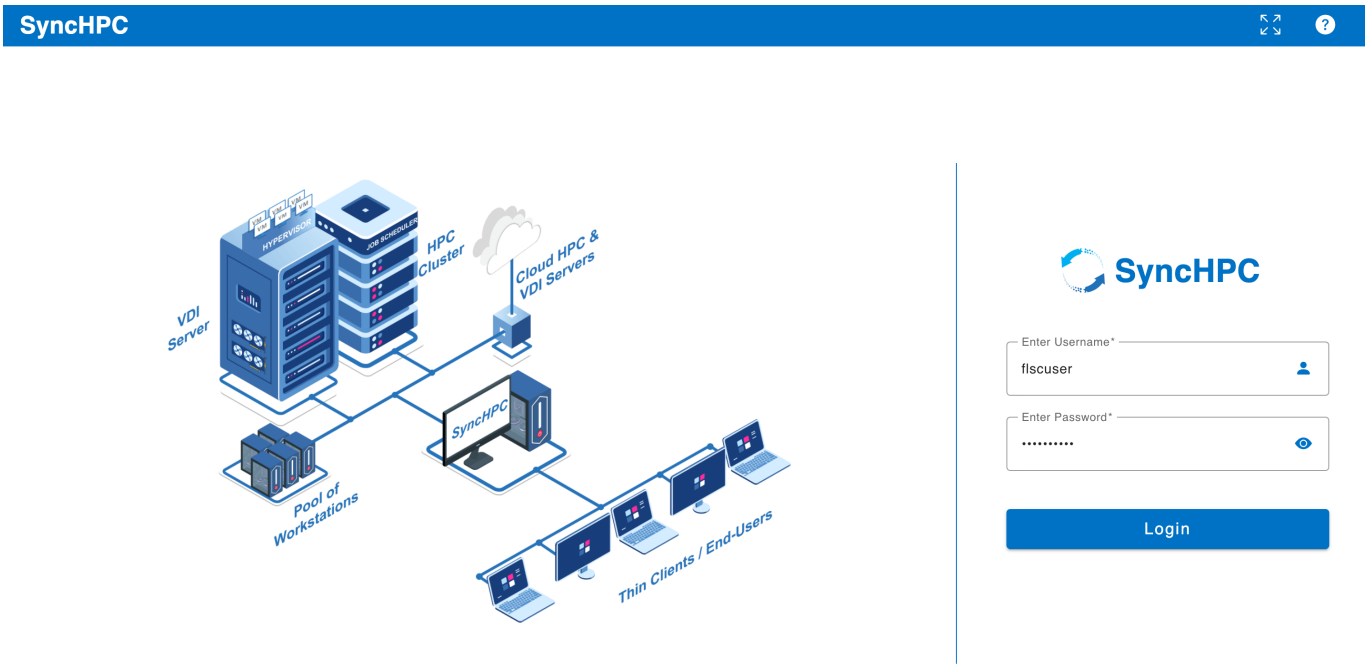
3. User Role Functionality

For making the user's daily activity an ease in 'SyncHPC', we have made the UI "user-friendly" and easy to understand so that the user does not need to access the backend for performing their tasks. Therefore, the "job Operator" user role has limited access to modules for submitting CAE jobs or requesting VDI sessions in "SyncHPC".

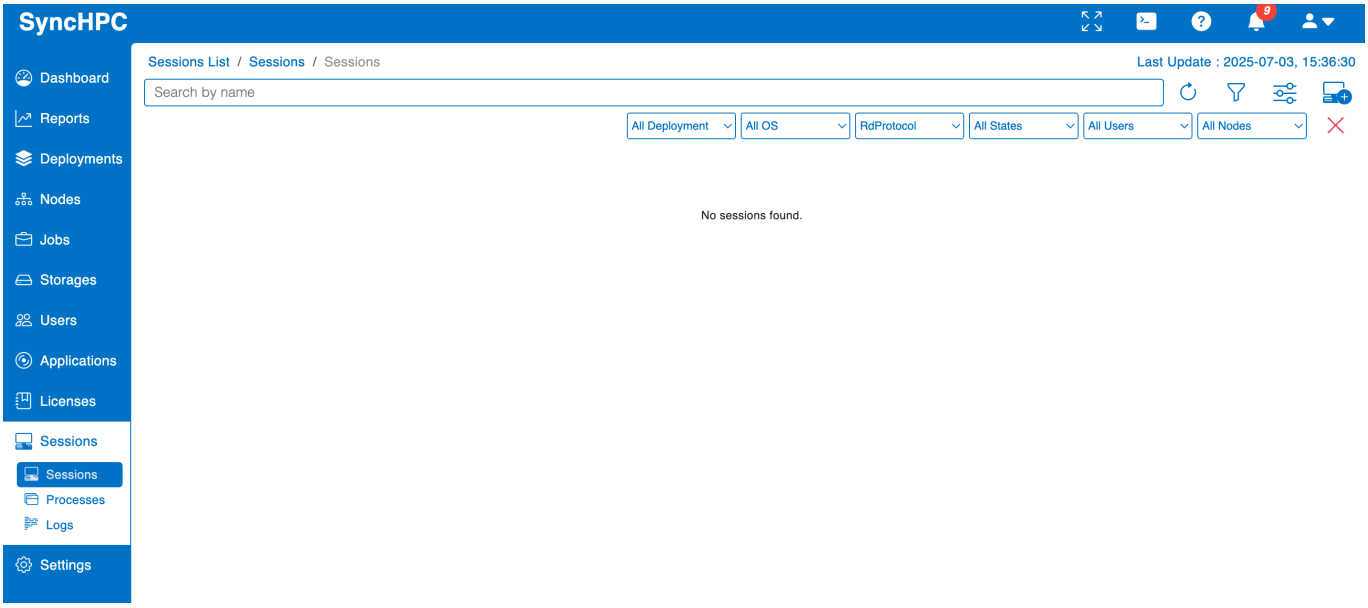
- Users can monitor the list of "Nodes" to cross-check whether resources are available before submitting jobs.
- In the Jobs module, users can submit jobs with the applications installed as per their requirements. Each application has different job forms with respective parameters to facilitate job submission.
- The storage module gives users access to create, manage, download, and upload their files in their respective user directories.
- Users can also monitor licenses with live data, keeping track of which licenses are being consumed by which user before submitting their jobs.
- The Sessions module is used for requesting and managing centralized virtual desktops for each user.

4. Requesting & Accessing SyncVDI Sessions

1. Login through the web browser using the SyncVDI link: https://hpc.cement1882.com/Session%20List/Sessions
2. Access SyncVDI through the domain and login with respective user credentials.



3. After logging in, users will be directed to the "session module" page, where they can access, manage, and request sessions with all the necessary features to facilitate their work.



User Interaction with Session Filters

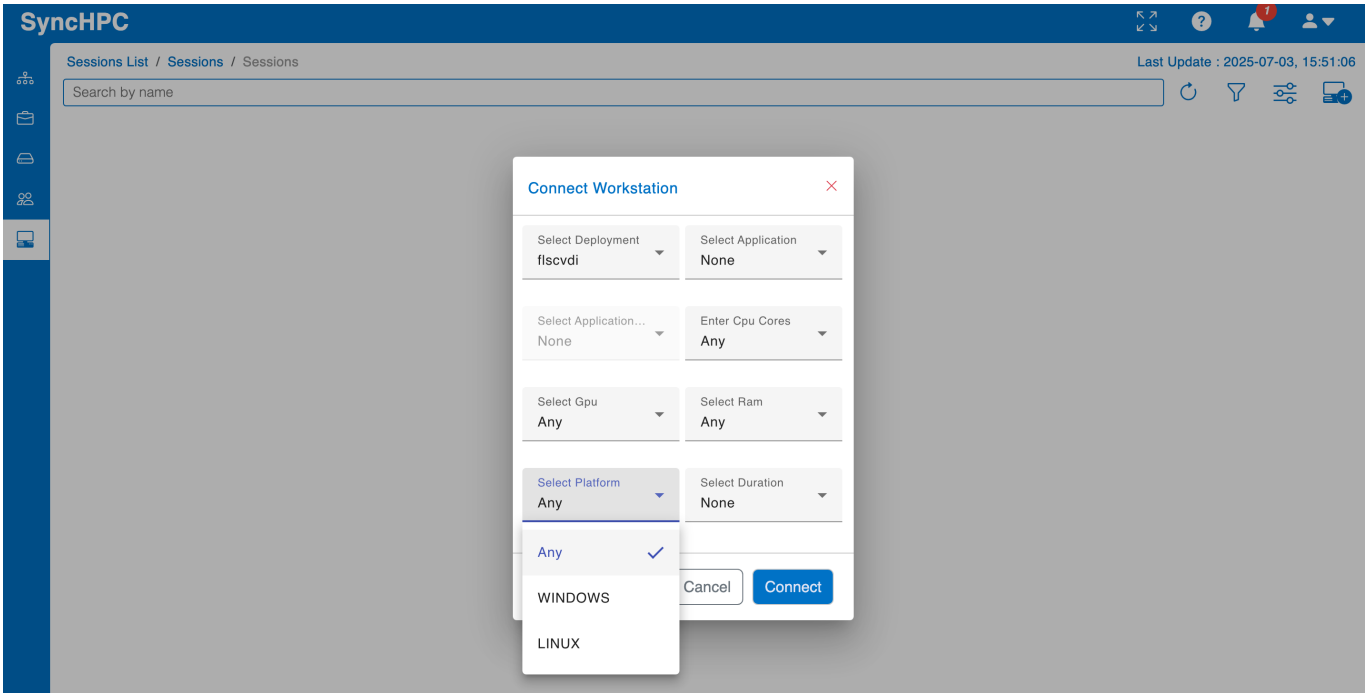
4. Users can interact with filters to sort their session listings based on various criteria:

- **All Deployments:** Displays the list of clusters to which the user is attached.
- **All OS:** Shows the list of sessions categorized as Windows/Linux.
- **All States:** Displays the state of sessions.
- **All Nodes:** Shows the current list of workstations requested or connected.

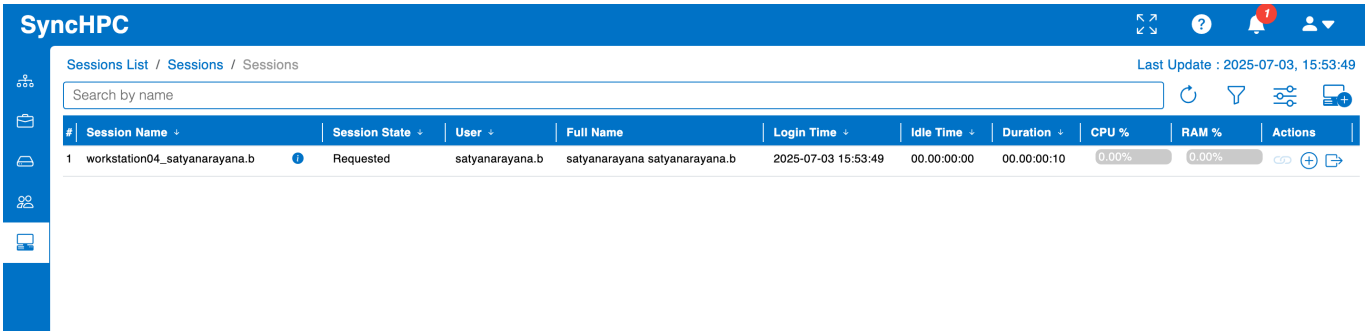
Logs Module

Users can access the Logs module located on the left-hand side of the page. This module contains records of sessions for tracking and monitoring purposes.

5. For requesting a new session, click on the request button which opens a pop-up box with configuration of session shown below. This pop-up window of requesting session can be configured with many aspects such as Selecting Deployments, Selecting Application, CPU cores, GPU, RAM, Platform, and Duration as per user requirements.

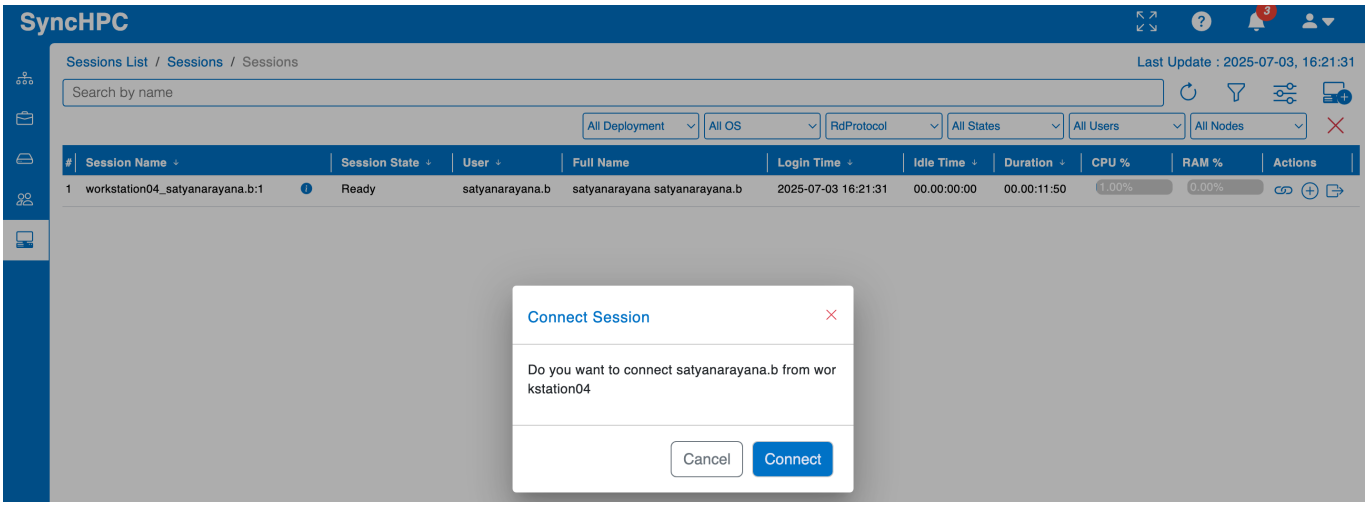


6. After selecting the configurations click on “Connect” button. A new session will be displayed on the screen with details of it. (shown in the picture below)

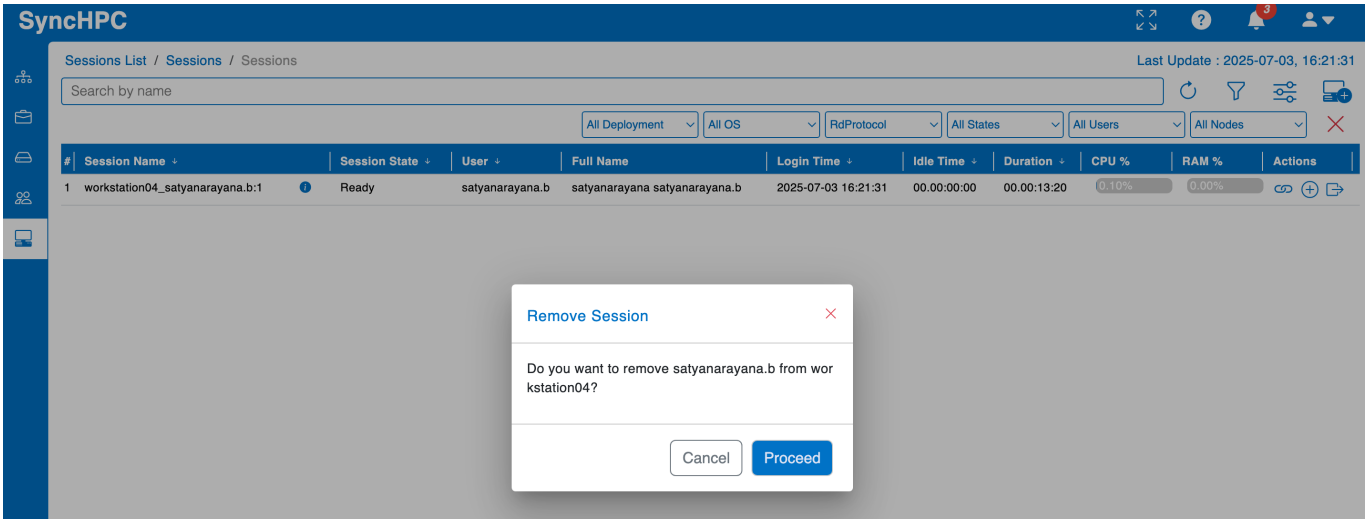


7. If the session was requested with any duration time, then the expiration time will also be shown with other details such as Login Time, GPU, CPU, and RAM percentages for monitoring purposes. The session Apps column shows which applications are being used on the workstation by users. All these header columns of the session can be edited with the “Column” button.

   User can search session by their name if multiple sessions are displayed in the listing.



8. User can add or remove the columns needed by selecting the checkboxes as shown.



9. For connecting or removing the sessions requested, user can access the clip-like button to connect the workstation and middle one with “+” sign for adding the time duration of session. The third button is used to remove the selected session.

   When clicking to the connect button, it downloads the RDP or .VNC files in the browser’s download section depending upon Window or Linux machine respectively. By accessing this file user can connect it to the machine. Please refer the below screenshots of the procedure to follow step by step.

   a. Requesting a Window and Linux Session. By selecting the Platform, user can choose to request a window or Linux machine.

Session States

b. When a user requests a session, the session consists of 4 different states which are:

- Requested
- Ready
- Active
- Disconnected

c. The workstation is assigned to a user until this the session remains in the requested state. However, the user can remove the session in this state or add duration but cannot connect it.

d. The machine is ready to connect in the current state which downloads the file for accessing the workstation.

e. When the user is currently using the workstation, it displays as active.

f. When the user disconnects the workstation, it displays as disconnected for a specific time frame which can be configurable from the admin side.

Ready Stage: In ready stage, workstations get assigned to user which can be connected via RDP or .VNC file

Active Stage: The Active stage shows which applications are currently being used on the workstation by users.

In Nodes module, user can see the workstations state as well like if the node is being used it will display as `busy`in the section.

If the workstations are available and active but not in used, it will be shown as above.

This document shows the step-by-step instructions for the user to get a detailed overview of requesting, managing, and removing sessions from the user perspective.

3. Troubleshooting

If you encounter any issues, contact our support team.

4. Support

For assistance or inquiries, contact our support team at `info@syncious.com` or visit our support portal